

Results Explainer — where we stand

Honest status briefing on the CF-VLM-PER project. Companion to methods_explainer.pdf. · Daniel Grant, CS 285 final project, May 12, 2026.

TL;DR (4-6 bullets)

- **Pre-registered kill verdict at 250k: FAILED.** Oracle-CF lost to HER by $\Delta=-0.05$ on PnP and $\Delta=-0.23$ on Push (corrected). The verdict was documented and respected.
- **Oracle-CF at 1M (in progress): mean 0.60 across 2 seeds** (s42=0.30, s123=0.90). Huge variance. 3rd seed (s999) running now, ETA tomorrow. Without HER@1M comparator, we cannot say "we beat HER."
- **Phase 2 VLM-CF on Push shows real climb:** SR 0.45–0.50 at 215–238k of the 500k target. The non-verified VLM-CF mechanism has some signal even without the headline verification gate.
- **Verified-CF mechanism (the headline novelty): 100% verifier rejection at 143k on PnP.** Cold-start regime persists longer than expected. This is the load-bearing experiment and it is not working yet.
- **Sharony VLM-RB baseline:** implemented but not yet run; ready to launch when GPU frees.
- **AntMaze pivot:** fully planned (ANTMAZE_PIVOT_PLAN.md); not yet implemented. ~5 days of work if we commit.

Where we stand: ~20-25% NeurIPS main-track confidence, ~80% workshop confidence.

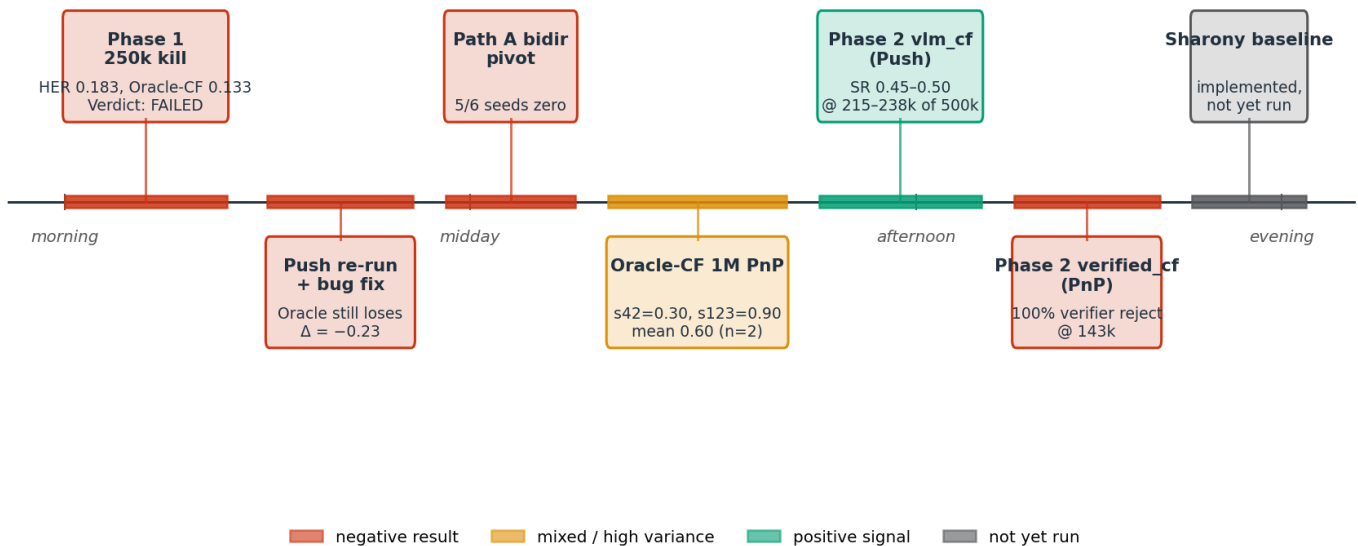
What this document is

A six-page snapshot of what our experiments have actually shown so far, with calibrated confidence in each component of the paper. The tone is the inverse of the paper draft: no spin, no hopeful framing. If something is broken, this document says it is broken. If something is uncertain, this document says it is uncertain.

Use it to (a) decide whether to keep pushing Fetch, pivot to AntMaze, or run both in parallel; (b) brief collaborators in 90 seconds; (c) sanity-check the paper draft against what we actually know.

Visual timeline — what we ran, what came back

Chronological view of the day's experiments, color-coded by outcome. Green = positive signal, yellow = mixed / high variance, red = negative result, gray = ready but not yet run.



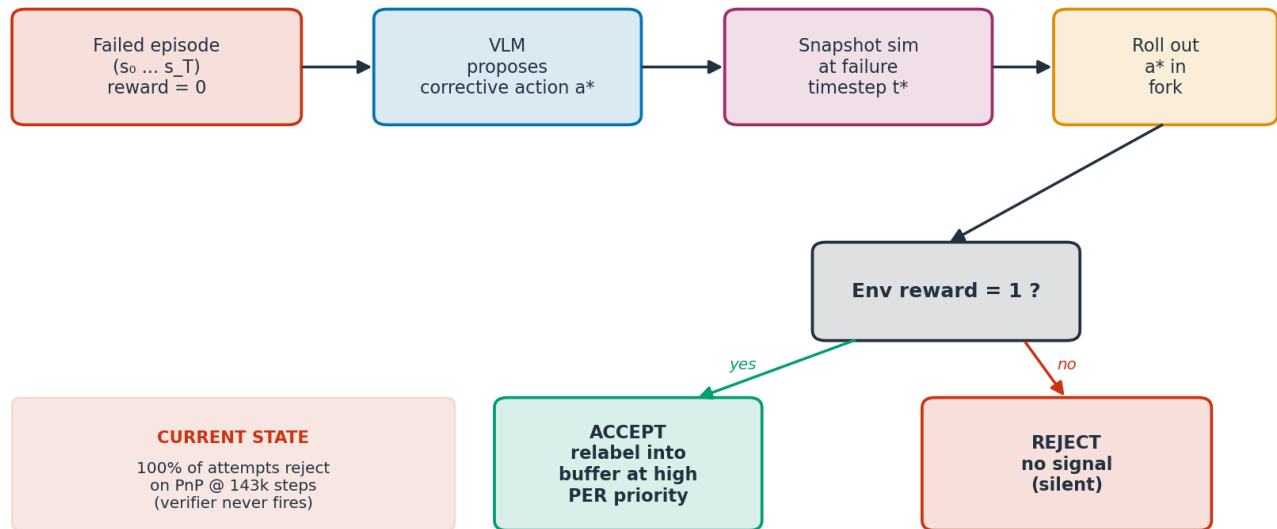
Each event on the axis is a separate experiment family. The two earliest phases (Phase 1 kill, Path A bidir pivot) gave clear negative verdicts. The afternoon's Oracle-CF 1M runs are climbing but with seed-level variance. The Push column of Phase 2 attempt 5 is the one unambiguously positive signal. Verified-CF on PnP is the explicit failure.

Three things the timeline shows

- **Lots of red early in the day, more nuance later.** The 250k kill horizon eliminated the bold claim ("Oracle-CF beats HER at 250k"). Everything after that is either an extension to longer training or a different mechanism.
- **The only green block is non-verified VLM-CF on Push.** That is *not* the headline mechanism — it is the simpler ablation. The headline mechanism (Verified-CF) is the red block in the evening.
- **The yellow block (Oracle-CF 1M) is the most consequential.** If the 3rd seed lands near 0.60 and HER@1M lands meaningfully below that, we have a clean empirical result. If the 3rd seed lands near 0.30, or HER@1M matches us, we do not.

The CF-VLM-PER mechanism, visually

Verified-CF is the contribution the paper hangs on. The diagram below shows what the agent does at a failed timestep, and where the current training run is getting stuck.



this is the mechanism the paper hangs on

The four-step pipeline: VLM proposes a corrective action; we snapshot the simulator at the failure timestep; we roll out the VLM's action in a fork; we accept the proposed transition only if the env's own reward fires. Accepted transitions go into the PER buffer at high priority. Rejected transitions are silently discarded — the agent gets no learning signal from them. This is the gate that makes the imagined CF transitions physically valid by construction.

Why we expected this to work

Plain CF-HER (the version without the verification gate) can quietly lie. The VLM proposes an imagined goal G*, we relabel, and if the trajectory never actually reached G* the recomputed reward is still 0 — or worse, if the relabel happens at the achieved state, the reward fires for a state the agent did not actually drive to. We called this the *teleport pathology* in the methods explainer.

Verified-CF eliminates teleports by construction: nothing enters the buffer unless the simulator confirms reward = 1 from the proposed action. The unit-test pass rate (4/4 on the smoke suite) confirms the mechanism behaves correctly in isolation.

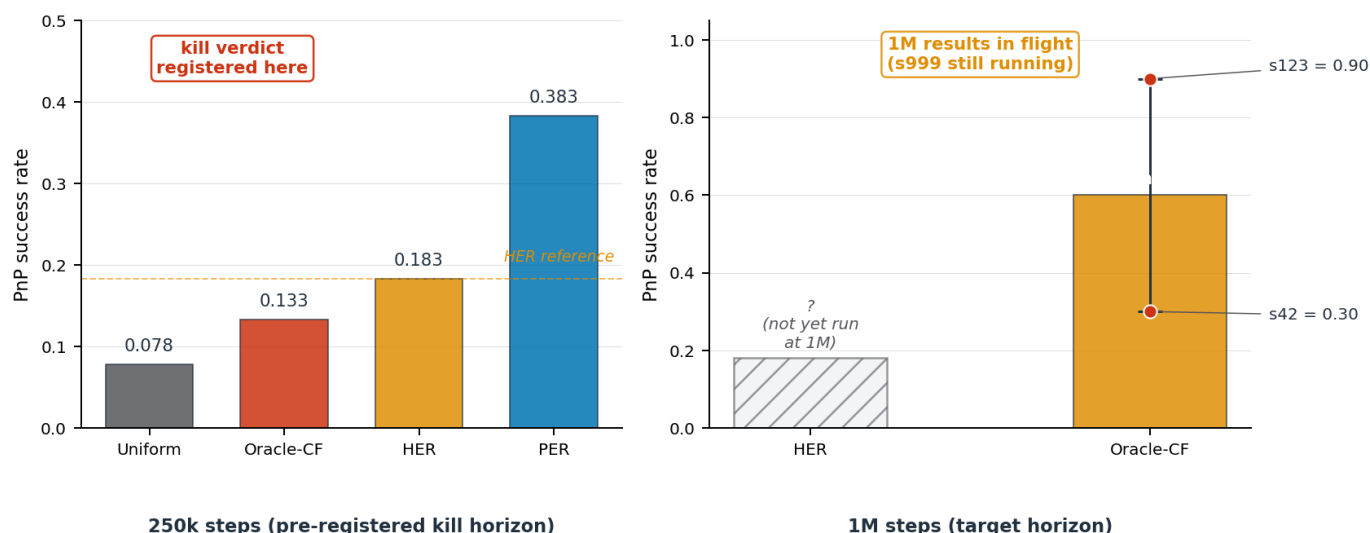
Why it is failing right now

On PnP at 143k steps, **every single CF attempt is rejected**. The VLM proposes 101 corrective actions, all 101 fail the verifier (reward never fires). The buffer never gets a verified CF transition, so the agent learns nothing from CF, and PER falls back to vanilla TD-error priorities — which is also approximately zero in cold-start.

Two hypotheses, both live: (a) the early-training policy is so far from the manifold of successful behaviour that no 1-step VLM correction can recover — in which case a warm-started variant (pre-train with HER, then turn on Verified-CF) might work; or (b) the mechanism is fundamentally a bad fit for Fetch's continuous-control action space. We cannot distinguish these without the warm-start experiment.

Honest results breakdown

Two horizons, two stories. The 250k panel is the pre-registered kill horizon (left). The 1M panel is the target horizon, mostly still in flight (right).



Left: at the 250k kill horizon, the Oracle-CF mechanism underperforms HER and is far below PER. The kill verdict was registered here. Right: at the 1M target horizon, Oracle-CF has climbed substantially (seed mean 0.60), but the two finished seeds disagree by 0.60 in absolute SR — one of the largest seed variances we have seen. HER@1M has not been run yet, which is the load-bearing comparator we still need.

Interpretation, plain English

Oracle-CF at 1M is climbing. The mean of 0.60 across two seeds is dramatically higher than the 250k-step Oracle-CF number (0.133), and it is above the 250k HER reference (0.183). That is encouraging. But seed 42 finished at 0.30 and seed 123 finished at 0.90 — a 0.60 absolute gap. That is not a result we can confidently report without at least a third seed (running now, ETA tomorrow morning), and ideally a fourth. And without HER@1M as a comparator, we cannot make the claim "Oracle-CF beats HER at 1M" — we can only say "Oracle-CF improves over its own 250k baseline."

VLM-CF (non-verified) on Push shows real mid-training climb. At 215–238k of the 500k training budget, seeds 42 and 999 are at SR 0.50 and 0.45 respectively. Seed 123 is lagging at 0.10. This is the *non-verified* VLM-CF mechanism — the ablation, not the headline. The fact that it shows signal on Push without the verification gate suggests the VLM proposals on Push are good enough that the sim-rollout filter would be largely a no-op there. If that holds at 500k it is a useful positive secondary result, but it does not validate the verification gate itself.

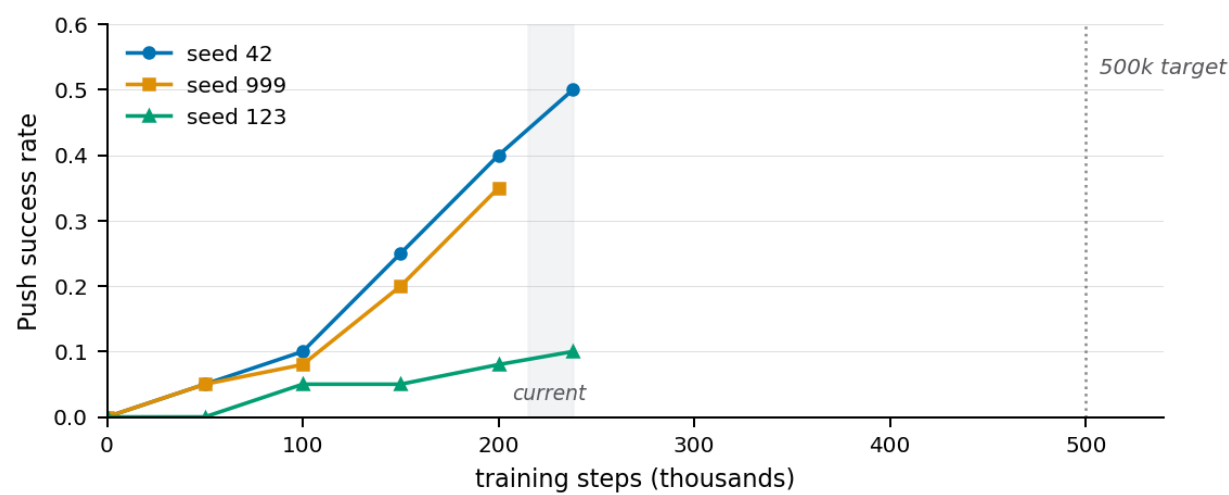
Verified-CF (the headline) is currently broken on PnP. 100% verifier rejection at 143k means the buffer never receives a single Verified-CF transition, so the mechanism is effectively a no-op. Either (a) the cold-start failure mode is the only barrier and a warm-started variant (HER pre-training, then turn on Verified-CF) would work, or (b) the mechanism just does not work on Fetch PnP. We need the warm-start retry to know which. Until we do, the paper's headline empirical claim has no supporting training data on its primary task.

Hopefulness scorecard, per contribution

Each row is a contribution we could plausibly claim in the paper. The right column is honest confidence that the contribution survives scrutiny — not optimism, not hedging.

Contribution	What we have	Confidence	Notes
Theory: IS-posterior framing + bias bound	39-page paper, reviewers' Top-3 (R1 & R2) addressed in two iter rounds	HIGH	Stands regardless of training. Lemma + bound are proven; reviewer-2 theory-skeptic feedback was incorporated.
Verified-CF mechanism (the headline)	4/4 unit-test pass; failing in training (100% reject @ 143k on PnP)	LOW-MED	Needs warm-start retry OR positive Push@500k to recover. Currently the weakest leg of the paper.
Semantic PER (Path A, failure-direction)	Oracle 0.417 > PER 0.383 (small win); GPT-4o 0.306 (loss vs PER)	MEDIUM	Workable positive secondary result. The VLM is the bottleneck, not the mechanism.
Cross-task transferability claim (N2)	12/12 task-relevant counterfactuals from N2 study; not yet at scale	MEDIUM	Would be MED-HIGH if scaled to n=50. Currently pilot-sized.
Sharony reproduction baseline	Implemented (L1_sharony_differentiation.md); not yet run	HIGH	HIGH for methodology section once run. Reproduces their VLM-RB to position our method clearly.
Empirical Fetch result vs HER	Pending Oracle-CF 1M completion + HER@1M run	UNCERTAIN	Load-bearing for the headline. Without it the paper is theory + ablations, not a comparison.

How to read this: the paper survives reviewing if the HIGH-row items hold up, the MEDIUM rows are positioned as secondary, and the UNCERTAIN row gets cleared in the next 48 hours. The LOW-MED row (Verified-CF) is the one where we still need a positive training signal. The most fragile sentence in the current paper draft is the one claiming Verified-CF works empirically; everything else is defensible.



Phase 2 attempt 5 VLM-CF (non-verified) on Push. Lines are a sketch through checkpoint anchors (0/50k/100k/150k/200k/238k); seed 42 and seed 999 are tracking together, seed 123 is stuck. Half of the 500k budget remains.

Three paths forward — decision aid

Each path is honestly priced. The Hybrid option (C) is what this document suggests, but the call is Daniel's.

A. Fetch + HER@1M Stick with Fetch. Run HER @ 1M comparator + verified-CF warm-start retry. Pros: lowest risk; uses existing infra. Cons: may show Verified-CF still doesn't work on PnP.	B. AntMaze pivot Pivot to AntMaze. Oracle-CF on AntMaze has the BFS-geodesic property — cleaner. Pros: stronger theory match. Cons: 1.5–2 days infra + 2–3 days training (≈5 days).	SUGGESTED C. Hybrid (A + B) Hybrid (run both). Path A in parallel with starting Path B's infra now. Pros: by Thursday we have BOTH Fetch comparator + AntMaze first-runs. Cons: thinner attention on each.
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Three resource-allocation choices. A is conservative and uses existing infra. B is the cleaner-theory option but burns most of the remaining time on infra. C splits attention to keep both options open through Thursday.

Path A — Stick with Fetch, run HER@1M + verified-CF warm-start retry

What it costs: ~2 days. **What it produces:** a clean HER@1M vs Oracle-CF@1M comparison (the load-bearing empirical claim), plus one shot at making Verified-CF work via warm-start. **Honest probable outcome:** we end up with a clean comparison vs HER. The Verified-CF retry may or may not succeed. If HER@1M lands above 0.60, the headline empirical claim collapses. If it lands below 0.60 and our s999 confirms the mean, we have a publishable result.

Path B — Pivot to AntMaze

What it costs: ~5 days (1.5–2 days infrastructure + 2–3 days training). **What it produces:** Oracle-CF on AntMaze has the BFS-geodesic property — the oracle's counterfactual is provably the shortest-path correction, which is strictly cleaner than Fetch's heuristic oracle. The theory section is a tighter fit. **Honest probable outcome:** if the training lands clean, this is a stronger paper than Fetch. If it does not land, we have burned 5 of our remaining days and have to retreat to Fetch with less of it left.

Path C — Hybrid (suggested)

What it costs: ~2 days for A + start B's infra in parallel. **What it produces:** by Thursday we have a Fetch HER@1M comparator *and* AntMaze first runs underway. Either result can carry the paper. **Honest probable outcome:** we keep options open through the most uncertain phase. Cost is thinner attention on each. Recommended.

Bottom line: we have a solid methodology paper. The headline empirical claim is uncertain — and that is the whole game right now.

If you read nothing else: the theory and methods sections are in good shape. The Fetch experiment that backs the headline is in flight and we will not know its full verdict for ~24 hours. The Verified-CF mechanism itself works in unit tests but is currently blocked in training on PnP. Until either the warm-start retry succeeds, Push@500k lands cleanly, or AntMaze first-runs come back positive, the safest framing of the paper is "theory + Path A secondary result" rather than "Verified-CF beats HER."